

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458453.9398 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.03 +/- 0.048
Transit depth (Rp/Rs)^2	0.087 +/- 0.28 [%]
Orbital Inclination (inc)	79.68 +/- 2.94
Airmass coefficient 1 (a1)	0.74 +/- 0.4
Airmass coefficient 2 (a2)	0.2 +/- 0.42
Scatter in the residuals of the lightcurve fit is	52.81 %
Best Comparison Star	None
Optimal Aperture	2.76
Optimal Annulus	3.0
Transit Duration (day)	0.026 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.03 $\pm$ 0.048 vs 0.1368 (deviation 2.23 σ)
Duration fit vs published	0.026 d vs 0.0758 d (deviation 1.72 σ)
Mid-time O-C	5.8 min
Depth SNR	0.6
Residual scatter	52.81 %

Light curve

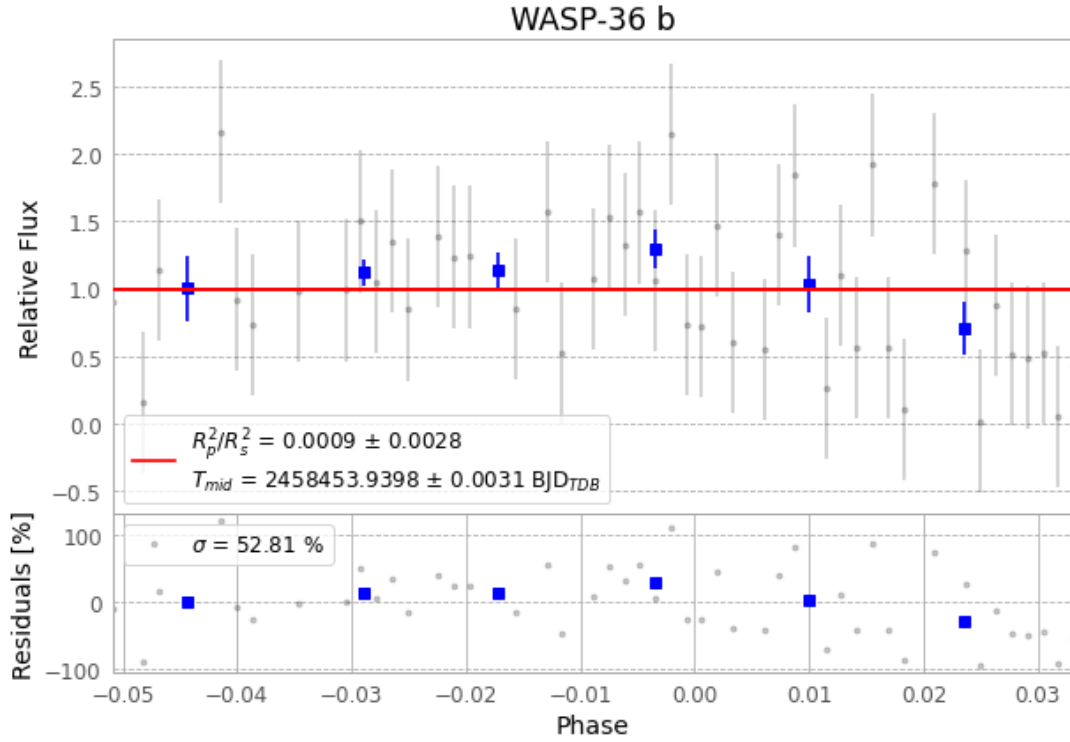


Figure 1 — FinalLightCurve_WASP-36 b_2018-12-01.png (SHA-256 15a75133288d90c1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [470, 393], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4a3551ea2b0342c0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

64 FITS frames (42 MB), WASP-36181201083612.FITS → WASP-36181201114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-181201.log (229205bd5964...), FinalLightCurve_WASP-36 b_2018-12-01.png (15a75133288d...), FinalLightCurve_WASP-36 b_2018-12-01.csv (fc27b7236972...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 22:02Z.